

Mathematical Derivation of Bistable Mechanisms for Vibration-Triggered Hook Deployment

Rolly Robot — Hook Bistable Actuator Simulations

Technical note for the Rolly hook-deployment simulations

2026

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1 Introduction

A *bistable* mechanical system has two stable equilibrium positions separated by an energy barrier. If it is not disturbed, it stays in whichever well it already occupies. Moving from one well to the other requires enough input energy to cross the intervening saddle; the rapid transition is the *snap-through* event [4, 6].

For the Rolly hook, the first well holds the hook retracted. A RobStride RS05 quasi-direct-drive motor drives the surrounding structure with a sinusoidal force or displacement. Near resonance, the hook motion grows until the relative coordinate crosses the saddle. After snap-through the second well holds the hook deployed without continuous motor power. The model here is therefore concerned with three quantities: the barrier height, the natural frequency about the retracted state, and the actuator amplitude needed to reach the saddle.

The physical model derived and simulated here is **Simulation 2 (Physical)**: a compliant curved-beam bistable mechanism following the geometry of Qiu et al. [4] and the 3D-printed Design 9 of Zolfagharian et al. [5], with a companion two-degree-of-freedom tuned mass damper (TMD) model [2].

1.1 Variable Map

The MATLAB scripts use compact variable names. The table below gives the mechanical meaning of the variables that drive the model.

Table 1: Plain-language map between MATLAB names and model variables.

MATLAB name	Symbol	Meaning
<code>h_apex</code>	h	Beam arch height at mid-span before snap. This sets the arch shape and the Q ratio.
<code>t_beam</code>	t	Beam thickness. A thicker beam lowers $Q = h/t$ but increases stiffness.
<code>l_span</code>	l	Hinge-to-hinge beam span.
<code>Q_beam</code> , Q	$Q = h/t$	Bistability ratio. The scripts now require $Q \geq 2.35$.
<code>n_cells</code>	N	Number of parallel bistable beams in the pack. Parallel beams share displacement and add force/stiffness.
<code>d_stroke_1cell</code>	$d_{f,1}$	Single-beam travel from retracted State 1 to deployed State 2.
<code>d_stroke</code>	d_f	Mechanism travel used by the ODE. In the parallel model this equals <code>d_stroke_1cell</code> .
<code>d_saddle</code>	d_s	Unstable saddle displacement. Crossing this displacement triggers snap-through.
<code>F_push_1cell</code>	$F_{\text{push},1}$	Peak force for one beam. The beam-pack force is $NF_{\text{push},1}$.
<code>A_coeff</code>	A	Cubic force coefficient in $F_{\text{restore}} = -Ad(d - d_s)(d - d_f)$.
<code>k_eff_s1</code>	$k_{\text{eff},1}$	Linearized stiffness near the retracted well. This sets $f_{n,1}$.
<code>m_hook</code>	m_h	Hook/reference mass attached to the bistable mechanism.
<code>M_plate</code>	M_P	Moving plate/structure mass used in the base-excitation model.
<code>xP</code>	x_P	Absolute plate displacement.
<code>xr</code> , <code>x_rel</code>	x_{rel}	Hook displacement relative to the plate. Snap occurs when this crosses d_s .
<code>A_min_c2</code>	A_{min}	Minimum prescribed plate amplitude for Config 2 snap in the linear estimate.

2 Compliant Curved-Beam Bistable Mechanism

2.1 Beam Geometry — Qiu Cosine Profile

Qiu et al. [4] proposed the initial (stress-free) shape of the curved beam as the first buckling mode of a fixed-end Euler beam, a cosine profile:

$$y_0(\xi) = \frac{h}{2} \left[1 - \cos\left(\frac{2\pi\xi}{l}\right) \right], \quad 0 \leq \xi \leq l \quad (1)$$

where h is the apex height [m] and l is the horizontal span [m]. The beam thickness is t [m], and the material elastic modulus is E [Pa].

Key geometric parameters:

$$Q = \frac{h}{t} \quad (\text{apex-height-to-thickness ratio}) \quad (2)$$

$$I = \frac{t^3}{12} \quad (\text{second moment of area per unit width}) \quad (3)$$

2.2 Bistability Condition

For a curved beam to exhibit bistability, two design conditions matter most [4, 5]:

1. **Q-ratio condition:** The apex height must substantially exceed the beam thickness:

$$Q = \frac{h}{t} \gtrsim 2.35 \quad (4)$$

Qiu et al. report that a second-mode-constrained curved beam becomes bistable once this geometric ratio is large enough; their analysis and force-displacement plots put the threshold at about $Q = 2.35$ [4]. The MATLAB model now enforces $Q \geq 2.35$ directly.

2. **Mode-2 suppression:** For a single beam, buckling mode 2 is asymmetric and prevents robust bistability. In the Zolfagharian Design 9, a *double-curved beam* connected by a central membrane at mid-span suppresses mode 2, so the snap-through path follows the symmetric constrained-beam behavior described by Qiu et al. [4].

2.3 Force-Displacement Relationship

2.3.1 Strain Energy Formulation

Using the Euler–Bernoulli beam assumptions, let the initial shape be $y_0(\xi)$ and the transverse displacement from that shape be $w(\xi)$, so $y(\xi) = y_0(\xi) - w(\xi)$ where w is the transverse displacement relative to the initial shape. The bending strain energy is then [4]:

$$U_b = \frac{EI}{2} \int_0^l \left(\frac{d^2w}{d\xi^2} \right)^2 d\xi \quad (5)$$

The axial compressive strain energy (due to shortening of the beam’s projected length) is:

$$U_a = \frac{EA}{2l} \left(\int_0^l \frac{dy_0}{d\xi} \frac{dw}{d\xi} d\xi \right)^2 \quad (6)$$

where $A = t \cdot b$ is the cross-sectional area (for width b).

The force-displacement relationship follows from virtual work: $F = \partial(U_b + U_a)/\partial d$, where d is the mid-point displacement (snapping direction).

2.3.2 Normalized Force-Displacement

Qiu et al. [4] show that the force-displacement curve depends only on Q and the normalized displacement $\bar{d} = d/h$. For the double-beam (mode-2 suppressed) configuration, the curve has the characteristic S-shape:

- **State 1** ($\bar{d} = 0$): zero force (equilibrium).
- **Peak push force** F_{push} at $\bar{d} \approx 0.27$: maximum force required to initiate snap-through.
- **Saddle** ($\bar{d} = \bar{d}_s$): force returns to zero (unstable equilibrium).
- **Valley force** $F_{\text{pop}} < 0$ at $\bar{d} \approx 0.82$: mechanism pulls itself toward State 2.
- **State 2** ($\bar{d} = 1$): zero force (second equilibrium).

2.3.3 Cubic Polynomial Fit

The simulation uses a cubic fit for the applied force in the compression-test convention. This is not a replacement for Qiu's modal solution or Zolfagharian's FEA; it is a compact surrogate with the correct equilibrium points and peak force:

$$F_{\text{ext}}(d) = A d (d - d_s) (d - d_f) \quad (7)$$

The coefficient A is calibrated so the force at the chosen peak location matches the Design 9 force level from Zolfagharian et al. [5]:

$$A = \frac{F_{\text{push}}}{d_{\text{push}} (d_{\text{push}} - d_s) (d_{\text{push}} - d_f)} \quad (8)$$

where $d_{\text{push}} \approx 0.27 d_f$ is the location of the force peak.

The *internal restoring force* on an attached mass (Newton's third law) is:

$$F_{\text{restore}}(d) = -F_{\text{ext}}(d) = -A d (d - d_s) (d - d_f) \quad (9)$$

2.4 Potential Energy (Curved-Beam Model)

Integrating the restoring force analytically:

$$\begin{aligned} V(d) &= - \int_0^d F_{\text{restore}}(\xi) d\xi = A \int_0^d \xi (\xi - d_s) (\xi - d_f) d\xi \\ &= A \int_0^d [\xi^3 - (d_s + d_f)\xi^2 + d_s d_f \xi] d\xi \\ &= A \left[\frac{d^4}{4} - \frac{(d_s + d_f)}{3} d^3 + \frac{d_s d_f}{2} d^2 \right] \end{aligned} \quad (10)$$

The energy barrier from State 1 to the saddle is:

$$\Delta E_1 = V(d_s) - V(0) = A \left[\frac{d_s^4}{4} - \frac{(d_s + d_f)}{3} d_s^3 + \frac{d_s d_f}{2} d_s^2 \right] \quad (11)$$

2.5 Natural Frequencies at Each Stable State

Linearizing $F_{\text{restore}}(d)$ about each equilibrium:

State 1 ($d = 0$):

$$k_{\text{eff},1} = -\left.\frac{dF_{\text{restore}}}{dd}\right|_{d=0} = A d_s d_f \quad (12)$$

$$\omega_{n,1} = \sqrt{\frac{k_{\text{eff},1}}{m}}, \quad f_{n,1} = \frac{1}{2\pi} \sqrt{\frac{A d_s d_f}{m}} \quad (13)$$

State 2 ($d = d_f$):

$$k_{\text{eff},2} = -\left.\frac{dF_{\text{restore}}}{dd}\right|_{d=d_f} = A d_f (d_f - d_s) \quad (14)$$

$$\omega_{n,2} = \sqrt{\frac{k_{\text{eff},2}}{m}}, \quad f_{n,2} = \frac{1}{2\pi} \sqrt{\frac{A d_f (d_f - d_s)}{m}} \quad (15)$$

Since $d_s < d_f$ by definition, and typically $d_s \approx 0.55 d_f$: $k_{\text{eff},2} = A d_f (d_f - d_s) \approx 0.45 A d_f^2$, while $k_{\text{eff},1} = A d_s d_f \approx 0.55 A d_f^2$, so $k_{\text{eff},2} < k_{\text{eff},1}$ and thus $f_{n,2} < f_{n,1}$.

2.6 Parallel Beam-Pack Mechanics

The robot layout is treated as a parallel pack of identical curved beams. The beams share the same hook displacement, so adding beams does *not* add stroke. Instead, the force and stiffness add:

$$d_{f,N} = d_{f,1} \quad (\text{same stroke}) \quad (16)$$

$$d_{s,N} = d_{s,1} \quad (\text{same saddle position}) \quad (17)$$

$$F_{\text{push},N} = N F_{\text{push},1} \quad (18)$$

The coefficient A for the beam pack therefore scales linearly:

$$A_N = N A_1 \quad (19)$$

because the displacement terms in the cubic polynomial are unchanged while the force is multiplied by N . The effective stiffnesses scale the same way:

$$k_{\text{eff},1,N} = A_N d_s d_f = N k_{\text{eff},1,1} \quad (20)$$

Thus adding parallel beams raises the natural frequency:

$$f_{n,1,N} = \sqrt{N} f_{n,1,1} \quad (21)$$

This correction keeps the saddle displacement on the order of a single beam's travel, not five times that travel.

2.7 1-DOF Hook Snap-Through Model

The hook mass m is connected to the fixed robot body through the bistable mechanism. External vibration $F_{\text{exc}}(t)$ is transmitted through the robot structure. The equation of motion is:

$$m\ddot{d} + c\dot{d} + F_{\text{restore}}(d) = F_{\text{exc}}(t) \quad (22)$$

where $c = 2\zeta m\omega_{n,1}$. The excitation is a linear chirp (frequency-modulated sinusoid):

$$F_{\text{exc}}(t) = F_0 \sin \left[2\pi \left(f_{\text{lo}} t + \frac{f_{\text{hi}} - f_{\text{lo}}}{2t_f} t^2 \right) \right] \quad (23)$$

where f_{lo} , f_{hi} are the chirp bounds [Hz] and t_f is the simulation duration [s]. As the instantaneous frequency passes through $f_{n,1}$, resonance amplifies the displacement until snap-through.

2.8 2-DOF Tuned Mass Damper Model

Following Zolfagharian et al. [5] (their Figure 4), the main structure mass m_1 is supported by spring k_{main} and driven harmonically. The TMD mass m_2 is attached to the same flexible ruler at distance L_2 from the fixed end, where the bistable mechanism repositions it.

The coupled equations of motion are:

$$m_1\ddot{x}_1 + c_{\text{main}}\dot{x}_1 + k_{\text{main}}x_1 + k_b(x_1 - x_2) + c_b(\dot{x}_1 - \dot{x}_2) = F_0 \sin(\omega t) \quad (24)$$

$$m_2\ddot{x}_2 - k_b(x_1 - x_2) - c_b(\dot{x}_1 - \dot{x}_2) = 0 \quad (25)$$

where k_b and c_b are the stiffness and damping of the bistable TMD connection, linearized at the current stable state.

2.8.1 Stiffness of the Flexible Ruler

The effective spring stiffness of a flexible cantilever beam (ruler) at distance L from the fixed support is [5]:

$$k = \frac{3EI_r}{L^3} \quad (26)$$

where E_r is the ruler's elastic modulus, I_r is its second moment of area. When the bistable mechanism shifts the TMD mass from L_1 to $L_2 = L_1 + d_f$:

$$k_{\text{TMD},2} = \frac{3E_r I_r}{L_2^3} = \frac{3E_r I_r}{(L_1 + d_f)^3} < k_{\text{TMD},1} \quad (27)$$

This reduction in TMD stiffness shifts the TMD natural frequency downward, changing the frequency at which it suppresses main-mass vibration.

2.8.2 Steady-State Frequency Response

In the frequency domain, the steady-state response of x_1 to $F_0 e^{i\omega t}$ is $X_1 e^{i\omega t}$. The impedance formulation gives:

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix} = \begin{Bmatrix} F_0 \\ 0 \end{Bmatrix} \quad (28)$$

where

$$Z_{11} = (k_{\text{main}} + k_b - m_1\omega^2) + i\omega(c_{\text{main}} + c_b) \quad (29)$$

$$Z_{12} = Z_{21} = -(k_b + i\omega c_b) \quad (30)$$

$$Z_{22} = (k_b - m_2\omega^2) + i\omega c_b \quad (31)$$

Solving by Cramer's rule:

$$X_1 = \frac{F_0 Z_{22}}{Z_{11}Z_{22} - Z_{12}^2} \quad (32)$$

The amplitude $|X_1(\omega)|$ vs. $\omega/(2\pi)$ [Hz] is plotted for three configurations: no TMD, TMD in State 1, TMD in State 2.

3 Base Excitation via Motor Torque Oscillation Through Shared Structure

3.1 Physical Setup

3.1.1 Mechanism component chain

The Zolfagharian Design 9 bistable mechanism consists of the following rigid bodies connected in series:

1. **Robot chassis / ground** — the fixed reference frame.
2. **Spring isolators** (k_s, c_s) — soft mounting elements (rubber mounts, vibration isolators) between the chassis and the plate assembly.
3. **Rigid plate assembly** (M_P) — one rigid body comprising the plate, the motor housing, and the *external blocks*. The external blocks are the side walls to which the bistable beam ends are pinned.
4. **Bistable curved beams** — the compliant Qiu cosine-profile beams spanning between the external blocks. These provide the nonlinear restoring force $F_{\text{restore}}(x_{\text{rel}})$ and potential energy $V(x_{\text{rel}})$.
5. **Hook** (m_h , also called the *reference mass*) — the deployable element attached at the beam apex. In State 1 the beam is arched and the hook is retracted; in State 2 the beam has snapped through and the hook is deployed.

The hook is connected to the plate assembly *only* through the bistable beams — there is no rigid link. When the plate assembly vibrates, the hook cannot follow instantaneously; the resulting **relative displacement** x_{rel} between the beam apex (hook) and the beam ends (external blocks on plate) is the quantity that drives snap-through.

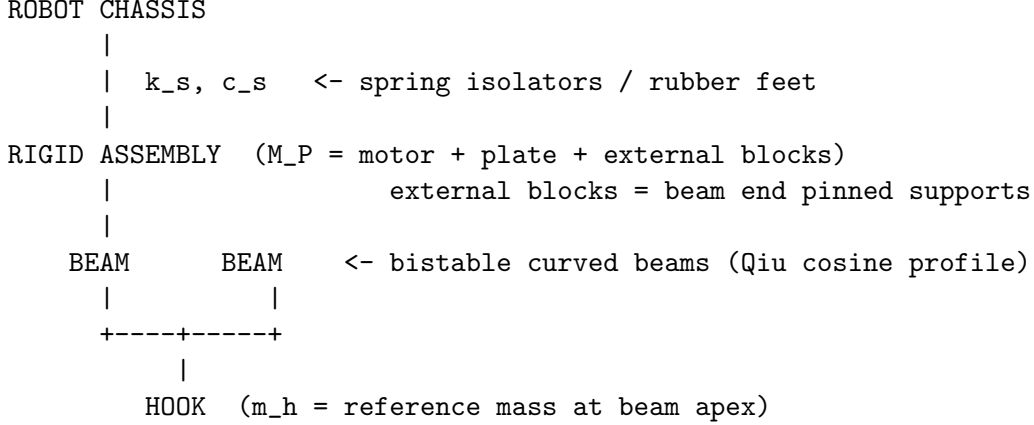


Figure 1: Physical component chain for the Zolfagharian bistable deployment mechanism.

3.1.2 Two excitation configurations

Configuration 1 — Motor on plate assembly. The motor housing is bolted directly to the plate (part of M_P). The motor shaft drives an eccentric mass or proof mass whose reaction force enters the plate through the bearing. The plate assembly vibrates; the hook responds via base excitation through the beams.

$$\text{Motor} \rightarrow \text{Plate} + \text{External blocks} \rightarrow \text{Beams} \rightarrow \text{Hook}$$

Configuration 2 — Motor off plate, drives via bearing. The motor housing is *not* part of the plate assembly. The motor output shaft drives the plate+external-blocks assembly directly through a bearing coupling (lever arm, crank, or shaft). In this configuration the motor can *prescribe* the plate displacement:

$$x_P(t) = A \sin(\omega_{n,1} t)$$

The hook responds to this base excitation through the beams. The motor housing mass is excluded from M_P .

$$\text{Motor} \rightarrow \text{Bearing} \rightarrow \text{Plate} + \text{External blocks} \rightarrow \text{Beams} \rightarrow \text{Hook}$$

Configuration 2 avoids the tuned-mass-damper anti-resonance problem because the motor enforces the plate amplitude regardless of the hook's dynamics.

3.1.3 Coordinates

Two coordinate systems are used throughout:

- $x_P(t)$: absolute displacement of the rigid plate assembly [m] (positive in the snap direction, i.e. from State 1 toward State 2)
 - $x_H(t)$: absolute hook displacement [m]
 - $x_{\text{rel}}(t) \equiv x_H(t) - x_P(t)$: hook displacement *relative to the plate assembly* [m] — this is the quantity that enters the bistable restoring force and potential energy
- Snap-through occurs when x_{rel} crosses the saddle point d_s with positive velocity.

3.2 Motor Torque Excitation Force

In torque/current control mode the RS05 applies a sinusoidal output torque $\tau_{\text{out}}(t)$. Through a coupling of moment arm r_{arm} [m] this becomes a linear force on the plate:

$$F_{\text{motor}}(t) = \frac{\tau_{\text{out}}}{r_{\text{arm}}} \sin(\omega_d t) \equiv F_0 \sin(\omega_d t) \quad (33)$$

where τ_{out} is the **measured output torque** read from the motor's torque-speed (T-N) curve at the instantaneous operating condition, and ω_d is the drive frequency set by the controller command. The amplitude F_0 is therefore *independent of frequency* — a key difference from a rotating-unbalance source whose amplitude grew as ω^2 .

For a linear chirp sweep from f_{lo} to f_{hi} over total time t_f , the instantaneous frequency and accumulated phase are:

$$f_{\text{inst}}(t) = f_{\text{lo}} + \frac{f_{\text{hi}} - f_{\text{lo}}}{t_f} t \quad (34)$$

$$\varphi(t) = 2\pi \left[f_{\text{lo}} t + \frac{f_{\text{hi}} - f_{\text{lo}}}{2t_f} t^2 \right] \quad (35)$$

The torque-driven force during a chirp is:

$$F_{\text{motor}}(t) = F_0 \sin(\varphi(t)) \quad (36)$$

The amplitude envelope F_0 is constant throughout the sweep, so the force delivered at $f_{n,1}$ equals the force at any other frequency in the band — the mechanism receives full excitation amplitude at every point in the chirp.

3.3 Gearbox Effects on Vibration Transmission

Any gearbox in the drivetrain between the motor rotor and the bistable plate acts as a mechanical filter. The gear ratio n_g amplifies the rotor's reflected inertia and creates a low-pass cut-off frequency above which vibrations are attenuated before reaching the plate. Two architectures relevant to bistable excitation are treated below.

3.3.1 Planetary Gearboxes

A planetary gearbox achieves its ratio n_g through a sun-planet-ring arrangement. Typical ratios range from 3:1 to about 20:1 per stage; multi-stage units reach 100:1. QDD motors use a single low-ratio planetary stage ($n_g \lesssim 10$) specifically to preserve backdrivability and broadband force transmission.

Reflected rotor inertia. At the output shaft, the effective inertia is:

$$J_{\text{eff}} = J_{\text{rotor}} n_g^2 + J_{\text{gears}} + J_{\text{output}} \quad (37)$$

The n_g^2 factor dominates: even a modest 5:1 ratio makes the rotor inertia appear $25\times$ larger at the output. Combined with the finite gear-mesh stiffness k_{gear} , this creates a mechanical low-pass cut-off frequency:

$$f_{\text{cut}} \approx \frac{1}{2\pi} \sqrt{\frac{k_{\text{gear}}}{J_{\text{eff}}}} \quad (38)$$

Vibrations at frequencies above f_{cut} are attenuated before reaching the plate; those well below f_{cut} pass through with little attenuation. A low gear ratio keeps f_{cut} high, so the motor remains substantially backdrivable and transmits a broad vibration band — a favorable property for this application.

Design rule: verify that $f_{n,1} < f_{\text{cut}}$ so that the bistable resonance lies in the pass-band of the actuator’s mechanical filter. This check should use the actual mounted motor and linkage geometry, not the bare motor datasheet value.

Gear-mesh frequency. The planet-gear tooth contacts produce a secondary vibration at:

$$f_{\text{mesh}} = n_g f_{\text{rotor}} N_{\text{teeth}} \quad (39)$$

where N_{teeth} is the planet tooth count. If f_{mesh} happens to coincide with $f_{n,1}$, this harmonic can contribute beneficially to snap-through excitation without any additional control effort. A low single-stage ratio keeps f_{mesh} well above the bistable resonance band, avoiding unintended parametric excitation.

3.3.2 High-Ratio Straight-Tooth Gearboxes

Multi-stage spur or helical gearboxes achieve ratios of $n_g = 20:1$ to $n_g > 100:1$ by cascading two or more stages of spur (straight-tooth) or helical gears. They are inexpensive and compact relative to harmonic or cycloidal alternatives at the same ratio, but carry two significant penalties for bistable vibration excitation.

Reflected inertia penalty. Because n_g is large, the n_g^2 term in eq. (37) dominates overwhelmingly. The cut-off frequency f_{cut} drops accordingly:

$$f_{\text{cut}} \propto \frac{1}{n_g} \quad (J_{\text{eff}} \approx J_{\text{rotor}} n_g^2) \quad (40)$$

At $n_g = 50$, a motor with $f_{\text{cut}} = 500$ Hz at $n_g = 1$ drops to roughly 10 Hz — potentially *below* the bistable resonance. Vibrations above this cut-off are mechanically filtered; the plate never receives the high-frequency force content needed for resonant snap-through.

Backlash dead-band. Multi-stage spur gearboxes accumulate backlash θ_{bl} (typically 5–30 arc-min total) from each mesh. Through the coupling lever arm r this creates a linear dead-band:

$$\delta_{bl} = \theta_{bl} r \quad (41)$$

During each oscillation half-cycle the drive must traverse δ_{bl} before force transmission resumes. The effective plate amplitude is reduced to:

$$A_{\text{eff}} = A_{\text{plate}} - \delta_{bl} \quad (42)$$

For snap-through, the minimum plate amplitude from eq. (95) must be met *net of backlash*: $A_{\text{eff}} \geq A_{\text{min}}$, i.e. $A_{\text{plate}} \geq 2\zeta_h d_s + \delta_{bl}$. If δ_{bl} is comparable to d_s , this constraint can become severe.

Gear-mesh noise. Multi-stage meshes generate harmonics at integer multiples of f_{mesh} (eq. (39)). If any harmonic coincides with $f_{n,1}$, it can excite unintended snap-through; this is harder to avoid in a multi-stage box than in a single-stage planetary.

Design rule. High-ratio straight-tooth gearboxes are suitable for bistable excitation only when $f_{n,1} \ll f_{\text{cut}}(n_g)$, the accumulated backlash $\delta_{bl} \ll d_s$, and the drive frequency is well below the first gear-mesh harmonic. In practice, these conditions are difficult to satisfy simultaneously at gear ratios above $\sim 20:1$; a low-ratio QDD or cycloidal stage is strongly preferred.

3.4 Two-Degree-of-Freedom Equations of Motion

The plate is modelled as a rigid body on a linear elastic support (stiffness k_s , damping c_s). The hook mass m_h is connected to the plate through the bistable element (restoring force $F_{\text{restore}}(x_{\text{rel}})$, eq. (9)) and a viscous damper c_h .

Plate equation of motion:

$$M_P \ddot{x}_P = F_{\text{motor}}(t) - \underbrace{k_s x_P - c_s \dot{x}_P}_{\text{support restraint}} - \underbrace{F_{\text{restore}}(x_{\text{rel}}) + c_h \dot{x}_{\text{rel}}}_{\text{reaction from hook}} \quad (43)$$

Hook equation of motion (absolute frame):

$$m_h \ddot{x}_H = F_{\text{restore}}(x_{\text{rel}}) - c_h \dot{x}_{\text{rel}} \quad (44)$$

The sign convention in eq. (43) reflects Newton's third law: the bistable spring pulls the plate with force $-F_{\text{restore}}$ (equal and opposite to what it exerts on the hook), and the dashpot exerts $+c_h \dot{x}_{\text{rel}}$ on the plate.

3.4.1 Relative-Coordinate Formulation

Subtracting eq. (43) (divided by M_P) from eq. (44) (divided by m_h):

$$\ddot{x}_{\text{rel}} = \ddot{x}_H - \ddot{x}_P = \underbrace{\frac{F_{\text{restore}} - c_h \dot{x}_{\text{rel}}}{m_h}}_{\ddot{x}_H} - \underbrace{\frac{F_{\text{motor}} - k_s x_P - c_s \dot{x}_P - F_{\text{restore}} + c_h \dot{x}_{\text{rel}}}{M_P}}_{\ddot{x}_P} \quad (45)$$

This is the governing equation for the bistable snap. The full state vector is $\mathbf{y} = [x_P, \dot{x}_P, x_{\text{rel}}, \dot{x}_{\text{rel}}]^\top$.

The support parameters are defined from the plate natural frequency f_P :

$$k_s = M_P (2\pi f_P)^2 \quad (46)$$

$$c_s = 2 \zeta_P M_P (2\pi f_P) \quad (47)$$

where ζ_P is the plate structural damping ratio.

3.5 Plate Transfer Function (Decoupled Analysis)

For small x_{rel} (early in the simulation, well before snap), the back-reaction of the bistable on the plate is negligible relative to the dominant terms F_{motor} , $k_s x_P$, and $c_s \dot{x}_P$. Under this approximation, the plate behaves as an independent SDOF oscillator:

$$M_P \ddot{x}_P + c_s \dot{x}_P + k_s x_P \approx F_{\text{motor}}(t) \quad (48)$$

Assuming harmonic excitation at frequency ω with constant amplitude F_0 , the steady-state plate displacement amplitude is:

$$|X_P(\omega)| = \frac{F_0}{\sqrt{(k_s - M_P \omega^2)^2 + (c_s \omega)^2}} \quad (49)$$

Substituting eqs. (46) and (47) and defining the frequency ratio $r_P = \omega/\omega_P$ where $\omega_P = 2\pi f_P$:

$$|X_P(\omega)| = \frac{F_0/(M_P \omega_P^2)}{\sqrt{(1 - r_P^2)^2 + (2\zeta_P r_P)^2}} = \frac{F_0/k_s}{\sqrt{(1 - r_P^2)^2 + (2\zeta_P r_P)^2}} \quad (50)$$

This is the standard frequency-response function of a driven SDOF oscillator. The amplitude peaks sharply at $r_P = 1$ (plate resonance, $\omega = \omega_P$), with the peak width governed by ζ_P .

3.6 Effective Excitation Force on the Bistable Hook

In the relative-coordinate frame, the plate's acceleration acts as an inertial (fictitious) force on the hook. The effective force amplitude at frequency ω is:

$$F_{\text{eff}}(\omega) = m_h \omega^2 |X_P(\omega)| \quad (51)$$

Substituting eq. (49):

$$F_{\text{eff}}(\omega) = \frac{m_h F_0 \omega^2}{\sqrt{(k_s - M_P \omega^2)^2 + (c_s \omega)^2}} \quad (52)$$

This is the force that the base excitation delivers to the bistable hook at frequency ω . Several observations follow directly:

1. The numerator scales as ω^2 (not ω^4 as in a rotating-unbalance source): the force still grows with frequency, but more gradually. At low frequencies the inertial coupling is weak; at high frequencies the hook is increasingly driven by the plate's acceleration.
2. The denominator contains the plate resonance denominator. When ω approaches ω_P , the denominator passes through a minimum, and F_{eff} passes through a maximum.
3. The maximum of F_{eff} occurs near (but not exactly at) ω_P . If ω_P is designed to coincide with $\omega_{n,1}$, the maximum excitation force is delivered precisely when the bistable hook is at its resonance — a double-resonance condition that amplifies the effective force by $Q_P \cdot Q_h$ relative to the quasi-static case.

3.7 General Snap-Through Condition

From eq. (91) (Optimization section), the minimum force required for snap-through at resonance $\omega = \omega_{n,1}$ is:

$$F_{0,\min} = 2\zeta_h k_{\text{eff},1} d_s = 2\zeta_h m_h \omega_{n,1}^2 d_s \quad (53)$$

Setting $F_{\text{eff}}(\omega_{n,1}) \geq F_{0,\min}$ and substituting eq. (52):

$$\frac{m_h F_0 \omega_{n,1}^2}{\sqrt{(k_s - M_P \omega_{n,1}^2)^2 + (c_s \omega_{n,1})^2}} \geq 2\zeta_h m_h \omega_{n,1}^2 d_s \quad (54)$$

Dividing both sides by $m_h \omega_{n,1}^2$ (positive):

$$\frac{F_0}{\sqrt{(k_s - M_P \omega_{n,1}^2)^2 + (c_s \omega_{n,1})^2}} \geq 2\zeta_h d_s \quad (55)$$

Rearranging, the **general snap condition** on the motor force amplitude is:

$$F_0 \geq 2\zeta_h d_s \sqrt{(k_s - M_P \omega_{n,1}^2)^2 + (c_s \omega_{n,1})^2} \quad (56)$$

The right-hand side is entirely determined by known system parameters. Notably, m_h has cancelled: the snap condition is *independent of hook mass*. However, $\omega_{n,1}$ itself depends on m_h through eq. (13), so the dependence is implicit.

3.8 Case 1: Tuned Plate ($f_P = f_{n,1}$)

3.8.1 Derivation

Set the plate support natural frequency equal to the bistable natural frequency:

$$f_P = f_{n,1} \iff \omega_P = \omega_{n,1} \iff k_s = M_P \omega_{n,1}^2 \quad (57)$$

Substituting into eq. (55). The term $k_s - M_P \omega_{n,1}^2$ vanishes exactly, leaving only the damping term in the denominator:

$$\sqrt{(M_P \omega_{n,1}^2 - M_P \omega_{n,1}^2)^2 + (c_s \omega_{n,1})^2} = c_s \omega_{n,1} \quad (58)$$

Substituting $c_s = 2\zeta_P M_P \omega_P = 2\zeta_P M_P \omega_{n,1}$:

$$c_s \omega_{n,1} = 2\zeta_P M_P \omega_{n,1}^2 \quad (59)$$

The snap condition eq. (55) becomes:

$$\frac{F_0}{2\zeta_P M_P \omega_{n,1}^2} \geq 2\zeta_h d_s \quad (60)$$

$$\boxed{F_0 \geq 4\zeta_h \zeta_P M_P \omega_{n,1}^2 d_s \quad (\text{Case 1: tuned plate})} \quad (61)$$

3.8.2 Physical Interpretation

Two points are useful for design:

1. **Scales with frequency.** Unlike a rotating-unbalance source, $F_0 \propto \omega_{n,1}^2$: a higher bistable resonance requires more actuator force. The available force should be verified against this threshold after any change to geometry or mass, using the actuator's rated output at the chosen coupling arm length.
2. **Doubly discounted by damping.** The factor $4\zeta_h \zeta_P$ means the threshold is $Q_P = 1/(2\zeta_P)$ times smaller than the off-resonance quasi-static baseline $2\zeta_h M_P \omega_{n,1}^2 d_s$. This Q_P factor is the mechanical quality factor of the plate resonance.

Physical explanation: at plate resonance the plate amplitude is amplified by Q_P relative to its quasi-static response; the hook then experiences Q_P times more inertial excitation, so Q_P times less actuator force is required.

3.8.3 Plate Amplitude at Snap Threshold

At the tuned threshold, from eq. (49) with the denominator reduced to $c_s \omega_{n,1}$:

$$|X_P|_{\text{tuned}} = \frac{F_0}{c_s \omega_{n,1}} = \frac{4\zeta_h \zeta_P M_P \omega_{n,1}^2 d_s}{2\zeta_P M_P \omega_{n,1}^2} = 2\zeta_h d_s \quad (62)$$

The plate displacement at threshold is $2\zeta_h d_s$. The hook displacement, amplified by $Q_h = 1/(2\zeta_h)$ at bistable resonance, reaches d_s :

$$|x_{\text{rel}}|_{\text{at snap}} = Q_h |X_P| = \frac{2\zeta_h d_s}{2\zeta_h} = d_s \quad (63)$$

as expected. The two resonances (plate at f_P , hook at $f_{n,1}$) in series provide amplification $Q_P \cdot Q_h = 1/(4\zeta_P \zeta_h)$ relative to the quasi-static case.

3.9 Case 2: Stiffness-Controlled Plate ($f_P > f_{n,1}$)

3.9.1 Regime Identification

When the plate support is stiffer than needed for resonance, $f_P > f_{n,1}$, the plate operates in the *stiffness-controlled* regime at the bistable's resonant frequency. Define the detuning ratio:

$$n \equiv \frac{f_P}{f_{n,1}} = \frac{\omega_P}{\omega_{n,1}} > 1 \quad (64)$$

In this regime $k_s > M_P \omega_{n,1}^2$, so the term $k_s - M_P \omega_{n,1}^2 = M_P(\omega_P^2 - \omega_{n,1}^2) > 0$ is the dominant contribution to the denominator of eq. (55).

3.9.2 Derivation

Substituting $k_s = M_P \omega_P^2 = M_P n^2 \omega_{n,1}^2$ and $c_s \omega_{n,1} = 2\zeta_P M_P \omega_P \omega_{n,1} = 2\zeta_P M_P n \omega_{n,1}^2$ into the denominator of eq. (56):

$$\begin{aligned} & \sqrt{(k_s - M_P \omega_{n,1}^2)^2 + (c_s \omega_{n,1})^2} \\ &= \sqrt{M_P^2 \omega_{n,1}^4 (n^2 - 1)^2 + M_P^2 \omega_{n,1}^4 (2\zeta_P n)^2} \\ &= M_P \omega_{n,1}^2 \sqrt{(n^2 - 1)^2 + (2\zeta_P n)^2} \end{aligned} \quad (65)$$

Substituting into eq. (56):

$$F_0 \geq 2\zeta_h M_P \omega_{n,1}^2 d_s \sqrt{(n^2 - 1)^2 + (2\zeta_P n)^2} \quad (\text{Case 2: } n = f_P/f_{n,1} > 1) \quad (66)$$

For light structural damping ($\zeta_P \ll 1$, so $2\zeta_P n \ll n^2 - 1$):

$$F_0 \approx 2\zeta_h M_P \omega_{n,1}^2 d_s (n^2 - 1) \quad (n > 1, \zeta_P \ll 1) \quad (67)$$

The threshold grows as $(n^2 - 1)$, which is approximately n^2 for large n . A plate resonance twice as high as the bistable resonance ($n = 2$) requires approximately $3\times$ the off-resonance inertia floor, and $75\times$ the tuned threshold.

3.9.3 Verification: Case 2 Reduces to Case 1 at $n = 1$

Setting $n = 1$ in eq. (66):

$$F_0 \geq 2\zeta_h M_P \omega_{n,1}^2 d_s \sqrt{0 + (2\zeta_P)^2} = 4\zeta_h \zeta_P M_P \omega_{n,1}^2 d_s \quad (68)$$

which recovers eq. (61).

3.10 Motor Mounted on the Bistable Plate: Coupling Model and Assumptions

This subsection addresses the specific physical configuration assumed throughout Section 3: the motor body is bolted directly to the same rigid plate that carries the bistable mechanism. It tracks the force path from an eccentric mass to the plate, shows why shaft angle is not a state variable in the hook dynamics, and states the assumptions explicitly.

3.10.1 Force Transmission Chain

When the motor shaft rotates, the eccentric mass m_e at radius e generates a centrifugal force. Before this force reaches the plate it passes through three mechanical interfaces, each with its own compliance:

1. **Shaft** (stiffness k_{sh} , damping c_{sh}): The shaft transmits the eccentric mass force to the inner bearing race. A rigid shaft ($k_{\text{sh}} \rightarrow \infty$) transmits 100%; a flexible shaft at speeds near its *critical speed* $\omega_{\text{crit}} = \sqrt{k_{\text{sh}}/m_e}$ can amplify or attenuate the transmitted force.
2. **Bearing** (stiffness k_b , damping c_b): The rolling-element bearing connects inner race to outer race (motor housing). The Hertzian contact stiffness of a typical deep-groove ball bearing is $k_b \sim 10^7\text{--}10^8$ N/m, far above the frequencies of interest here, so bearing deflection is negligible.
3. **Motor mount bolts and plate**: The motor housing is assumed rigidly bolted to the plate. Any compliance in this joint (bolt stretch, washer deformation) is folded into the effective bearing stiffness.

The *bearing transmission function* $T_b(\omega)$ quantifies how much of the raw centrifugal force $m_e e \omega^2$ actually reaches the plate at frequency ω :

$$T_b(\omega) = \frac{k_b + i c_b \omega}{(k_b - m_e \omega^2) + i c_b \omega} \quad (69)$$

The force entering the plate is therefore:

$$F_{\text{plate}}(\omega) = T_b(\omega) \cdot m_e e \omega^2 \quad (70)$$

Rigid-bearing limit. When $k_b \gg m_e \omega^2$ (bearing far from its own resonance):

$$T_b(\omega) \rightarrow 1 \quad (k_b \rightarrow \infty) \quad (71)$$

All force transfers to the plate. Rolling-element bearings typically have Hertzian contact stiffnesses $k_b \sim 10^7\text{--}10^8$ N/m. As long as $m_e \omega^2 \ll k_b$ — satisfied at any bistable resonance frequency well below the bearing's own resonance — $T_b \approx 1$ and the full centrifugal force reaches the plate, justifying $F_{\text{motor}} = m_e e \omega^2 \sin(\omega t)$.

3.10.2 Single-Axis Projection of the 2-D Rotating Force

The centrifugal force is a vector that rotates in the plane of the motor shaft:

$$\mathbf{F}_{\text{cent}}(t) = m_e e \omega^2 [\cos(\omega t) \hat{x} + \sin(\omega t) \hat{y}] \quad (72)$$

Only the component aligned with the bistable stroke direction (take $\hat{y}$ as the snap axis) does useful work on the bistable:

$$F_{\text{snap}}(t) = \mathbf{F}_{\text{cent}}(t) \cdot \hat{y} = m_e e \omega^2 \sin(\omega t) \quad (73)$$

The orthogonal component $m_e e \omega^2 \cos(\omega t)$ loads the plate transversely; it is transmitted to the plate support as a lateral force but does not couple into the bistable snap direction and is neglected in the 1-DOF model.

Practical implication. Aligning the motor shaft so that the eccentricity plane contains the snap direction maximises the effective force. Misalignment by angle α reduces F_{snap} by $\cos \alpha$.

3.10.3 Motor Shaft Angle Does Not Appear in the Hook Dynamics

A key property of the rotating-unbalance architecture is that the *motor shaft angle* $\theta(t) = \omega t + \theta_0$ does not appear in the bistable hook's equation of motion — only the resulting plate acceleration does.

Proof. The hook EOM in relative coordinates is (eq. (45)):

$$m_h \ddot{x}_{\text{rel}} = F_{\text{restore}}(x_{\text{rel}}) - c_h \dot{x}_{\text{rel}} - m_h \ddot{x}_P(t) \quad (74)$$

The right-hand side contains $\ddot{x}_P(t)$ — the plate acceleration. This quantity depends on $F_{\text{snap}}(t) = m_e e \omega^2 \sin(\omega t)$ through the plate dynamics, but the shaft angle θ itself never appears. Whether the motor has rotated 100 turns or 1000 turns is irrelevant; only the instantaneous force amplitude and frequency matter.

Contrast with direct drive. In a crank-slider or lever-arm direct-drive arrangement, the motor shaft angle θ sets the hook position geometrically: $x_{\text{hook}} = f(\theta)$. A change in θ immediately moves the hook. With rotating unbalance there is no such kinematic constraint: the hook position is determined entirely by the energy balance between the inertial excitation and the bistable restoring force. The motor can spin freely at constant speed without tracking or position control.

3.10.4 Summary of Model Assumptions

Table 2: Assumptions in the shared-plate rotating-unbalance model and their validity for the Rolly deployment mechanism.

Assumption	Mathematical form	Validity / break-down
Rigid bearing	$T_b(\omega) = 1$	Valid: $k_b \gg m_e \omega^2$ near the current resonance
Rigid motor shaft	$\omega_{\text{crit}} \gg \omega$	Valid if shaft critical speed $\gg f_{n,1}$
Constant motor speed	$\dot{\omega} = 0$	Valid if motor torque bandwidth $\gg f_{n,1}$
Single-axis force	$F = m_e e \omega^2 \sin(\omega t)$	Loses factor $\cos \alpha$ if shaft mis-aligned
Rigid motor-to-plate joint	$k_{\text{joint}} \rightarrow \infty$	Valid for bolted flange; invalid for press-fit
Plate as rigid body	No plate bending modes	Valid if plate structural modes $\gg f_{n,1}$
Linear plate support	k_s, c_s constant	Invalid near plate support nonlinearity

3.11 Motor Housing on the Plate with a Proof Mass on the Output Shaft

This subsection treats a distinct configuration: the motor housing is bolted to the bistable plate as before, but instead of a continuously spinning eccentric mass, the motor output shaft drives a *proof mass* M_{pm} that **oscillates linearly** along the snap axis relative to the plate. This is sometimes called a *reaction-mass actuator* or *proof-mass actuator* and is used when independent control of force amplitude and frequency is required.

3.11.1 Equations of Motion

Define coordinates:

- $x_P(t)$: absolute plate displacement (positive in snap direction)
- $\delta(t) = x_{pm}(t) - x_P(t)$: proof mass displacement *relative* to plate
- $x_{\text{rel}}(t)$: hook displacement relative to plate (unchanged from eq. (45))

The motor applies an internal force $F_{\text{act}}(t)$ between the plate (housing) and the proof mass. By Newton's third law the plate receives $-F_{\text{act}}$ and the proof mass receives $+F_{\text{act}}$.

Proof mass (absolute frame):

$$M_{pm} \ddot{x}_{pm} = M_{pm} (\ddot{x}_P + \ddot{\delta}) = F_{\text{act}}(t) \quad (75)$$

Plate (absolute frame):

$$M_P \ddot{x}_P = -F_{\text{act}}(t) - k_s x_P - c_s \dot{x}_P - F_{\text{restore}}(x_{\text{rel}}) + c_h \dot{x}_{\text{rel}} \quad (76)$$

Adding eqs. (75) and (76) to eliminate F_{act} :

$$(M_P + M_{pm})\ddot{x}_P + M_{pm}\ddot{\delta} = -k_s x_P - c_s \dot{x}_P - F_{\text{restore}}(x_{\text{rel}}) + c_h \dot{x}_{\text{rel}} \quad (77)$$

The key observation is that F_{act} has been eliminated. The plate equation is now driven entirely by the proof mass *relative acceleration* $\ddot{\delta}$.

3.11.2 Prescribed Harmonic Proof-Mass Motion

If the motor drives the proof mass to oscillate at amplitude Δ and frequency ω :

$$\delta(t) = \Delta \sin(\omega t) \quad \implies \quad \ddot{\delta}(t) = -\Delta \omega^2 \sin(\omega t) \quad (78)$$

Substituting into eq. (77):

$$(M_P + M_{pm})\ddot{x}_P = \underbrace{M_{pm} \Delta \omega^2 \sin(\omega t)}_{\text{effective driving force}} - k_s x_P - c_s \dot{x}_P - F_{\text{restore}}(x_{\text{rel}}) + c_h \dot{x}_{\text{rel}} \quad (79)$$

This has *exactly the same form* as the rotating-unbalance plate equation (eq. (43)), with two substitutions:

$$m_e e \longrightarrow M_{pm} \Delta \quad (\text{force product}) \quad (80)$$

$$M_P \longrightarrow M_P + M_{pm} \quad (\text{effective plate mass}) \quad (81)$$

The plate-transfer and snap-threshold results above carry over with these substitutions.

3.11.3 Snap Condition for the Proof-Mass Configuration

Applying the same derivation as section 3.8 with substitutions eqs. (80) and (81):

Tuned plate ($f_P = f_{n,1}$):

$$\boxed{M_{pm} \Delta \geq 4 \zeta_h \zeta_P (M_P + M_{pm}) d_s} \quad (82)$$

Defining the proof-mass ratio $\mu = M_{pm}/M_P$ and rearranging:

$$\Delta \geq 4 \zeta_h \zeta_P M_P d_s \frac{1 + \mu}{\mu} = \frac{4 \zeta_h \zeta_P M_P d_s}{\mu} + 4 \zeta_h \zeta_P d_s \quad (83)$$

As $\mu \rightarrow \infty$ (proof mass much heavier than plate), the required stroke $\Delta \rightarrow 4 \zeta_h \zeta_P d_s$, independent of M_P . As $\mu \rightarrow 0$ (very light proof mass), $\Delta \rightarrow \infty$: a tiny mass must oscillate with enormous stroke to produce enough force.

3.11.4 Comparison with Rotating Unbalance

Table 3: Proof-mass oscillator vs. rotating unbalance: key differences for the same bistable plate.

Property	Rotating unbalance	Proof-mass oscillator
Force amplitude	$m_e e \omega^2$ (fixed by geometry)	$M_{pm} \Delta \omega^2$ (Δ controllable)
Force–frequency	Always grows as ω^2	Grows as ω^2 unless $\Delta(\omega)$ is adjusted
Effective plate mass	M_P	$M_P + M_{pm}$
Motor direction	Continuous rotation (one direction)	Reciprocating (reverses each half-cycle)
Snap product	$m_e e \geq 4\zeta_h \zeta_P M_P d_s$	$M_{pm} \Delta \geq 4\zeta_h \zeta_P (M_P + M_{pm}) d_s$
Shaft angle in EOM?	No	No (same decoupling argument)

The proof-mass oscillator is strictly more flexible than the rotating unbalance because Δ can be varied in real time via motor current. The cost is that the motor must reverse direction each cycle rather than spinning continuously, and the heavier effective plate mass ($M_P + M_{pm}$) raises the snap threshold.

3.11.5 Proof-Mass Resonance

If the shaft connecting motor to proof mass has compliance k_{pm} , the proof mass forms a secondary resonance:

$$\omega_{pm} = \sqrt{\frac{k_{pm}}{M_{pm}}} \quad (84)$$

Driving near ω_{pm} amplifies Δ by $Q_{pm} = 1/(2\zeta_{pm})$, reducing the required motor force by the same factor. This creates a *triple resonance* path: plate resonance at ω_P , bistable hook resonance at $\omega_{n,1}$, and proof-mass resonance at ω_{pm} . Tuning all three to the same frequency gives amplification $Q_P \cdot Q_h \cdot Q_{pm}$ but also narrows the bandwidth and makes the system sensitive to parameter variation.

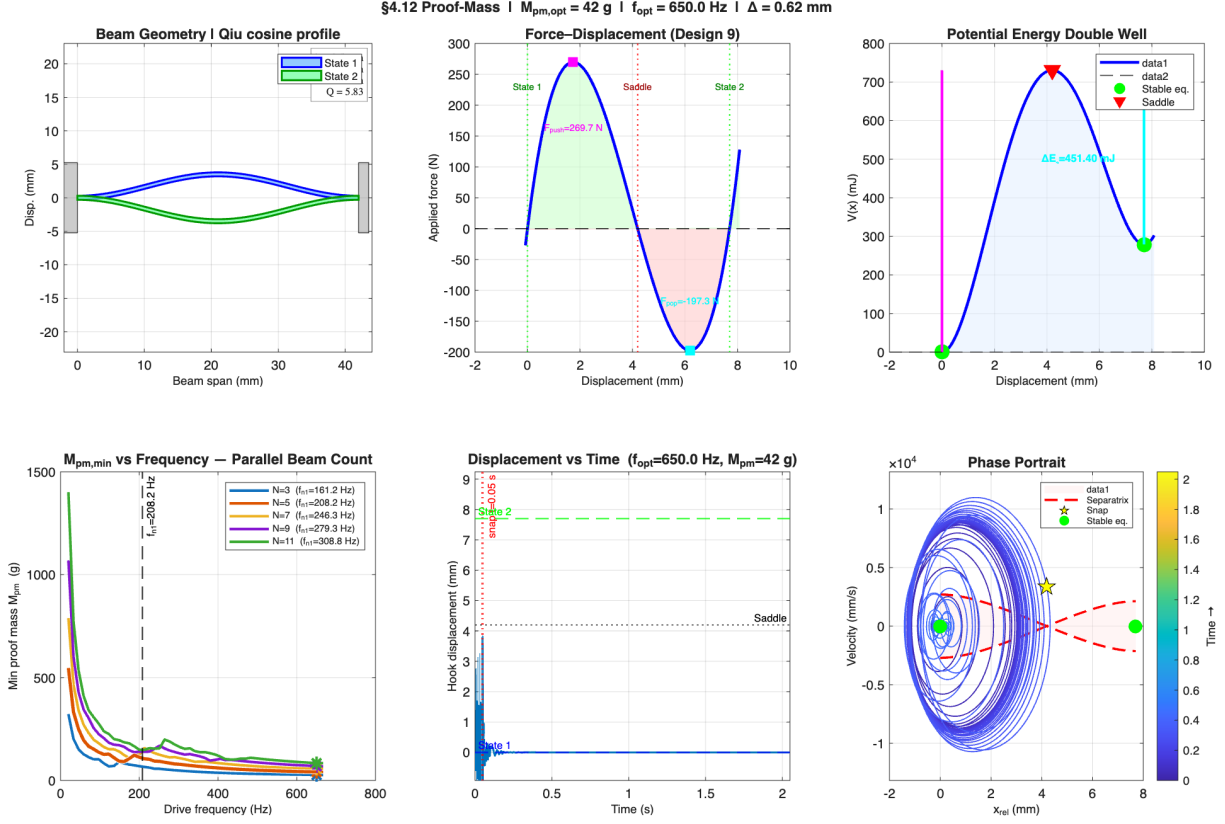


Figure 2: Proof-mass actuation analysis: minimum proof mass $M_{pm,min}$ and corresponding snap-through drive frequency as a function of design parameters (N parallel beams, hook mass m_h).

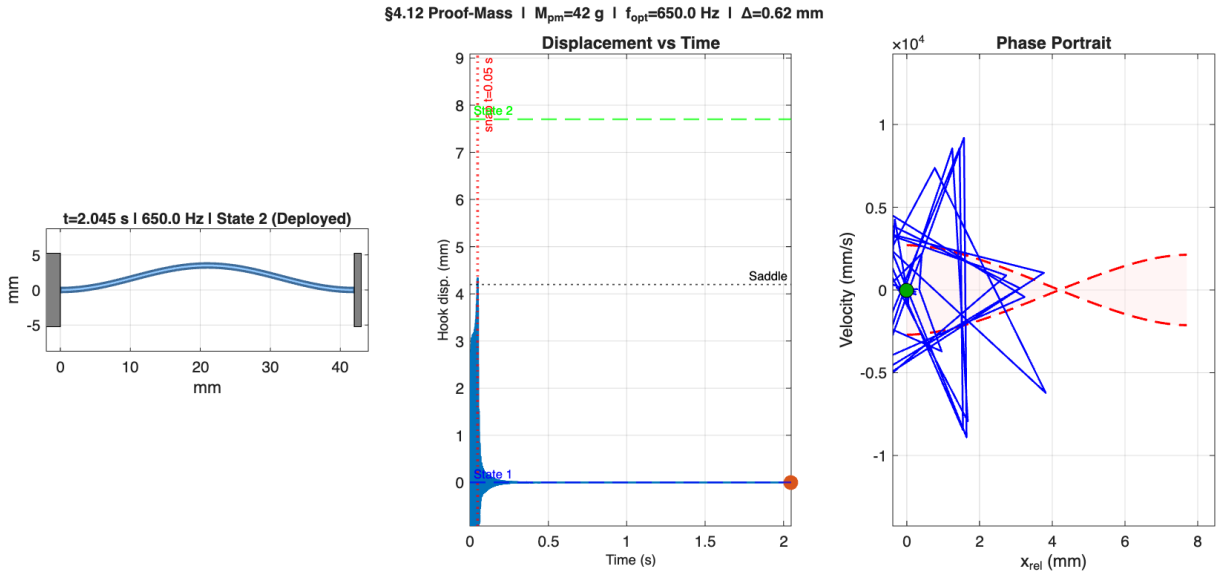


Figure 3: Animation snapshot for the Section 3.11 proof-mass actuator at the optimal proof-mass parameter. The proof mass oscillates on the motor output shaft, driving the plate and hook into snap-through.

4 Actuator Limit Used in the Simulation

The detailed RS05 motor parameters and CAN command notes have been moved to the separate RS05 notes (`rs05_motor_specifications.tex` and `rs05_motor_specifications.typ`). The main derivation only needs the output-shaft velocity limit used by the proof-mass and prescribed-motion models:

$$\omega_{\max} = 50.3 \text{ rad/s.} \quad (85)$$

For sinusoidal shaft motion at frequency f and angular amplitude Θ ,

$$v_{\text{peak}} = \Theta 2\pi f \leq \omega_{\max} \quad (86)$$

so the maximum angular amplitude is

$$\Theta_{\max}(f) = \frac{\omega_{\max}}{2\pi f} \quad (87)$$

and the corresponding linear stroke through lever arm r is

$$\delta_{\max}(f) = \frac{\omega_{\max} r}{2\pi f}. \quad (88)$$

For snap-through, the available stroke must reach the saddle d_s , so

$$r_{\min}(f) = \frac{2\pi f d_s}{\omega_{\max}}. \quad (89)$$

5 Optimization

5.1 Snap-Through Threshold

From linear resonance theory, the steady-state hook amplitude at $\omega = \omega_{n,1}$ is:

$$A_{\text{ss}} = \frac{F_0}{2\zeta_h k_{\text{eff},1}} \quad (90)$$

Snap-through occurs when $A_{\text{ss}} \geq d_s$, giving:

$$\boxed{F_{0,\min} = 2\zeta_h k_{\text{eff},1} d_s} \quad (91)$$

This is the *minimum force applied directly at the hook*. For base excitation (motor on plate), the motor force required is larger by a factor that depends on the plate dynamics and drive frequency.

5.2 Mass Actuation Force

The central design question for the Rolly deployment mechanism is: *for a given mass budget added to the robot, what is the maximum force that mass can deliver to the bistable hook?*

5.2.1 Proof-Mass Oscillator (Section 3.11)

A proof mass M_{pm} oscillating at frequency ω and amplitude Δ delivers inertial force $F = M_{pm} \Delta \omega^2$. At the velocity limit (eq. (87)):

$$\Delta_{\max}(\omega) = \frac{\omega_{\max} r}{\omega} \quad (92)$$

The maximum force per unit proof mass is:

$$\frac{F_{\max}}{M_{pm}} = \Delta_{\max} \omega^2 = \omega_{\max} r \omega \quad (93)$$

This grows *linearly with drive frequency*. Higher frequency delivers more force for the same proof mass — up to the limit imposed by the bistable dynamics (TMD anti-resonance at $\omega_{n,1}$; see Section 3.8). The coupled relative-mode resonance at ω_{rel} is the correct drive frequency:

$$\omega_{\text{rel}} = \sqrt{\frac{k_{\text{eff},1}}{M_{\text{red}}}}, \quad M_{\text{red}} = \frac{m_h M_P}{m_h + M_P} \quad (94)$$

The minimum proof mass to snap at ω_{rel} is found by binary search over the nonlinear ODE.

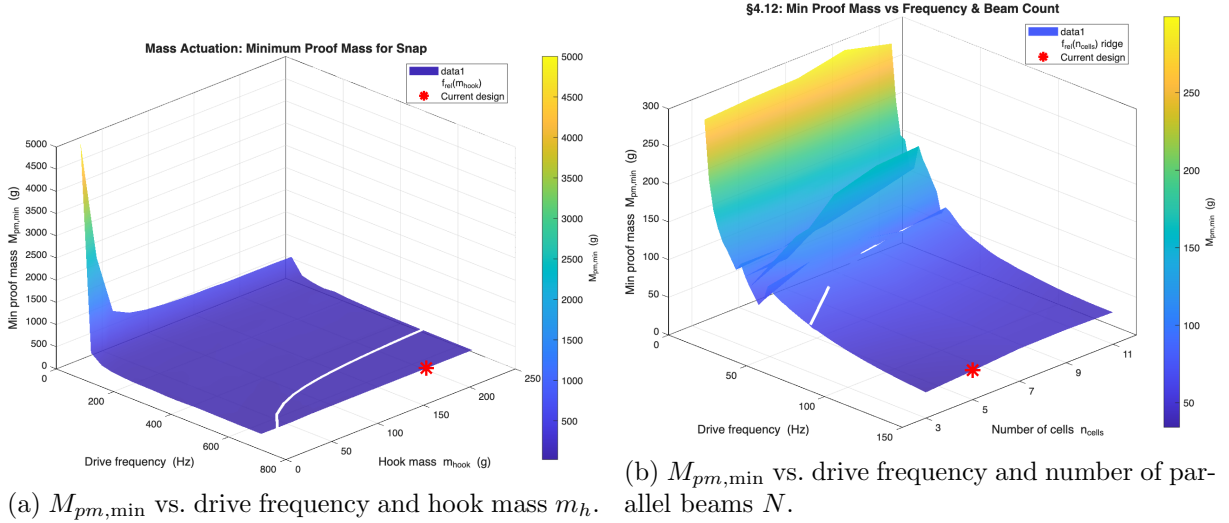


Figure 4: 3-D design surfaces for the minimum required proof mass $M_{pm,min}$ to achieve snap-through, swept over the drive frequency and two key design variables. Each surface is obtained by binary search over the nonlinear ODE at every grid point.

5.2.2 Config 2: Prescribed Base Motion (no added mass)

The motor prescribes the plate displacement $x_P(t) = A \sin(\omega_{n,1}t)$. The minimum plate amplitude for resonant snap is:

$$A_{\min} = 2 \zeta_h d_s \quad (95)$$

The required motor torque at resonance ($\omega = \omega_{n,1}$):

$$\tau = m_h A_{\min} \omega_{n,1}^2 r = 2 \zeta_h k_{\text{eff},1} d_s r \quad (96)$$

This is *independent of hook mass m_h* . No additional hardware mass is required — the motor already present for locomotion is used.

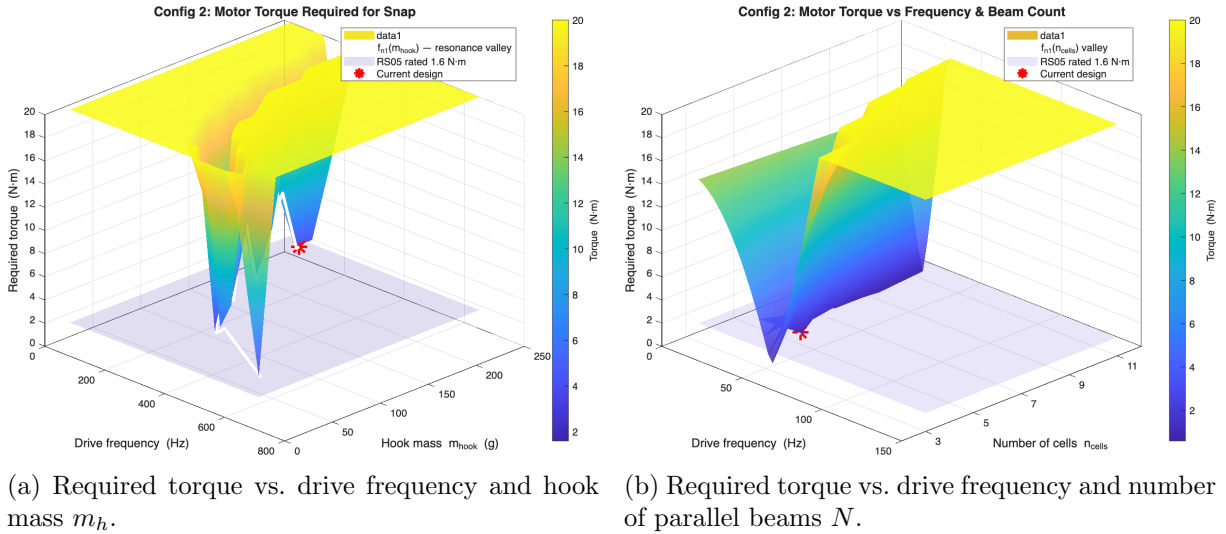


Figure 5: 3-D design surfaces for the required motor torque under Configuration 2 (prescribed base motion) for snap-through, swept over the drive frequency and two key design variables.

5.3 Parallel Beam-Count Scaling (N beams)

For the current model, N is the number of parallel bistable beams. The beams share the same hook displacement, so adding beams increases force and stiffness, not stroke:

$$d_f = d_{f,1} \quad (\text{stroke unchanged}) \quad (97)$$

$$d_s = d_{s,1} \quad (\text{saddle unchanged}) \quad (98)$$

$$k_{\text{eff},1} \propto N \quad (\text{stiffness rises}) \quad (99)$$

$$f_{n,1} \propto \sqrt{N} \quad (\text{natural frequency rises}) \quad (100)$$

$$F_{0,\text{min}} \propto N \quad (\text{force threshold rises}) \quad (101)$$

The old series assumption made the saddle displacement look too large. In the parallel model, $N = 5$ raises the stored energy and stiffness while keeping the hook travel at the single-beam value.

The post-snap potential energy available to accelerate the hook to State 2 is

$$\Delta E_2 = V(d_s) - V(d_f). \quad (102)$$

It grows with N because the force-displacement curve scales upward, not because the stroke grows.

5.4 Impact Force at State 2

When the hook arrives at $x_{\text{rel}} = d_f$ it has kinetic energy accumulated from two sources:

1. **Kinetic energy at saddle crossing** KE_{saddle} (from resonant build-up).
2. **Potential energy released** by the bistable spring from saddle to State 2: ΔE_2 .

The hook velocity at impact (ignoring damping over the short snap transit):

$$v_{\text{impact}} = \sqrt{v_{\text{saddle}}^2 + \frac{2 \Delta E_2}{m_h}} \quad (103)$$

At the minimum snap threshold $v_{\text{saddle}} \approx 0$, so $v_{\text{impact}} \approx \sqrt{2 \Delta E_2 / m_h}$.

The peak force on the target object (Hertz contact model):

$$F_{\text{peak}} = v_{\text{impact}} \sqrt{k_{\text{contact}} m_h} \quad (104)$$

where k_{contact} is the contact stiffness. The impact force scales as $\sqrt{k_{\text{contact}}}$: steel-on-steel ($k \sim 10^{10}$ N/m) gives $\sim 3000\times$ more peak force than rubber-on-rubber ($k \sim 10^4$ N/m) for the same impact velocity.

The impact momentum (relevant for deforming targets):

$$p = m_h v_{\text{impact}} = \sqrt{2 m_h \Delta E_2} \quad (105)$$

grows as $\sqrt{m_h}$ — heavier hooks deliver more momentum.

6 Nomenclature

Table 4: Symbol definitions used in this document and the MATLAB simulations.

Symbol	Unit	Description	Typical value
<i>Simulation 2 — Curved-beam bistable</i>			
E	Pa	Elastic modulus (ABS)	2300×10^6
t	m	Beam thickness	0.6×10^{-3}
h	m	Beam apex height	3.5×10^{-3}
l	m	Beam span	42×10^{-3}
Q	—	Apex-height-to-thickness ratio	h/t (target $\gtrsim 2.35$)
N	—	Number of parallel beams	5
d_f	m	Mechanism stroke	7.70×10^{-3}
d_s	m	Saddle position	$0.545 d_f$
A	N/m ³	Cubic polynomial coefficient	from eq. (8)
F_{push}	N	Peak push force (beam pack)	$N \times 52.4$ (current model)
F_{pop}	N	Valley force	from cubic fit
$k_{\text{eff},1}$	N/m	Stiffness at State 1	$A d_s d_f$
$k_{\text{eff},2}$	N/m	Stiffness at State 2	$A d_f (d_f - d_s)$
$f_{n,1}$	Hz	Natural freq. at State 1	$\omega_{n,1}/(2\pi)$
$f_{n,2}$	Hz	Natural freq. at State 2	$\omega_{n,2}/(2\pi)$
m_1	kg	Main mass (TMD model)	0.150
m_2	kg	TMD mass	0.01875

continued on next page

Symbol	Unit	Description	Typical value
k_{main}	N/m	Main spring stiffness	7253 (35 Hz)
E_r	Pa	Ruler elastic modulus	210×10^9
<i>Section 3 — Base excitation model</i>			
x_P	m	Absolute plate displacement	(simulated)
x_{rel}	m	Hook displacement relative to plate	(simulated)
M_P	kg	Plate + bistable assembly mass (motor excluded)	0.200
f_P	Hz	Plate support natural frequency	set to $f_{n,1}$
ζ_P	—	Plate structural damping ratio	0.02
k_s	N/m	Plate support stiffness	$M_P(2\pi f_P)^2$
c_s	N·s/m	Plate support damping	$2\zeta_P M_P(2\pi f_P)$
F_0	N	Motor force amplitude at plate	$\tau_{\text{out}}/r_{\text{arm}}$
τ_{out}	N·m	Measured RS05 output torque (from T-N curve)	≤ 5.5
r_{arm}	m	Coupling moment arm	0.010
n_g	—	RS05 planetary gear ratio	7.75
J_{eff}	kg·m ²	Reflected rotor inertia at output	$J_{\text{rotor}} n_g^2$
f_{cut}	Hz	Gearbox low-pass cut-off frequency	$\frac{1}{2\pi} \sqrt{k_{\text{gear}}/J_{\text{eff}}}$
f_{mesh}	Hz	Gear-mesh frequency	$n_g f_{\text{rotor}} N_{\text{teeth}}$
n	—	Detuning ratio $f_P/f_{n,1}$	1 (tuned)
Q_P	—	Plate quality factor $1/(2\zeta_P)$	25
Q_h	—	Hook quality factor $1/(2\zeta_h)$	10
<i>Section 5 — Optimization and mass actuation</i>			
N	—	Number of parallel bistable beams	5
f_{rel}	Hz	Coupled relative-mode resonance $= \sqrt{k_{\text{eff},1}/M_{\text{red}}}/(2\pi)$	22
M_{red}	kg	Reduced mass $= m_h M_P/(m_h + M_P)$	0.133
M_{pm}	kg	Proof mass (oscillating on motor shaft)	optimised
Δ	m	Proof-mass oscillation amplitude	velocity-limited
ω_{max}	rad/s	RS05 max shaft angular velocity (480 RPM)	50.3
$M_{pm,\text{min}}$	kg	Minimum proof mass to snap at given (f, N)	from sweep
A_{min}	m	Minimum plate amplitude for Config 2 snap	$2\zeta_h d_s$
τ_{C2}	N·m	Config 2 motor torque $= 2\zeta_h k_{\text{eff}} d_s r$	2.87
ΔE_2	J	Post-snap potential energy released (saddle to State 2)	grows with N
v_{impact}	m/s	Hook velocity at State 2 impact	$\sqrt{2\Delta E_2/m_h}$
k_{contact}	N/m	Target contact stiffness (Hertz)	$10^4\text{--}10^{10}$
F_{peak}	N	Peak impact force $= v_{\text{impact}} \sqrt{k_c m_h}$	—

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